

I/O pin that requires an external pullup. The ALERT pin is an output that can be used as an external interrupt to the host processor if certain application conditions are detected.

Charge Control

Lithium-ion/polymer batteries are very common in a wide variety of portable electronic devices because they have a very high energy density, minimal memory effect, and low self-discharge. However, care must be taken to avoid overheating or overcharging these batteries to prevent damage to the batteries potentially resulting in dangerous outcomes/explosive results. By operating in safe temperature ranges, at safe voltages, and under safe current levels, the overall safety of the lithium-ion/polymer batteries can be assured throughout the life of the battery.

MAX17335 controls the charging voltage and current dynamically based on the JEITA charge profile, step-charging, battery temperature, and temperature of the charging FET. The charge current is reduced at low battery voltage (prequal), low and high temperature, or when the charging FET is in a temperature or power dissipation limit. [Figure 1](#) shows the typical charge profile through the operating range of a battery. MAX17335 has several regulation and control options, shown in the list below.

- **Autonomous Charger** with non-volatile configuration.
- **Constant Current Regulation.** Configurable from 185mA to 2331mA (with a 10mΩ sense resistor) and step size configurable as described in [Table 75](#), with 1% accuracy. Scalable with a sense resistor for larger or smaller currents and batteries. See [Table 13](#).
- **Constant Voltage Regulation.** Configurable in 5mV steps from 3.56V to 4.835V (or 10mV steps from 2.42V to 4.97V) with 0.25% accuracy.
- **Constant Power Regulation.** The MAX17335 measures the pack and battery voltage and charging current to calculate the power in the FET and sense resistor. The IC regulates heat with a configurable threshold (scalable with a sense resistor).
- **Temperature Regulation.** The IC regulates the FET temperature to a configurable threshold.
- **Supplement Mode** quickly supported with ideal diode.
- **Parallel Cell Management**, including cross-charge blocking.
- **Manual Charge Control.** If enabled, the host can manually override the autonomous control and write the ChargingVoltage and ChargingCurrent registers directly.
- **Age Scheduled Charge Duration.** If enabled, the MAX17335 can derate the ChargingVoltage and ChargingCurrent as the battery ages. See [nAgeChgCfg](#).
- **Zero Volt Charging/Blocking** and Battery Prequalification.

[Table 1](#) and [Table 2](#) show an example of the charge profile with step charging and JEITA profile changing the target charge current and charge voltage. See [nIChgCfg1](#), [nIChgCfg2](#), [nVChgCfg1](#), [nVChgCfg2](#), [nStepCurr](#), [nStepVolt](#), [nTPrtTh1](#), [nTPrtTh2](#) and [nDesignCap](#) for configuration details.

Table 1. Charging Current with Step Charging and JEITA

TEMPERATURE	TOO COLD	COLD2	COLD1	COOL	ROOM	WARM	HOT1	HOT2	TOO HOT
	<-10°C	-10°C – 0°C	0°C – 10°C	10°C – -20°C	20°C – 40°C	40°C – 45°C	45°C – 50°C	50°C – 55°C	>55°C
Step 4	No Charging	0.438C	0.484C	0.531C	0.609C	0.547C	0.500C	0.453C	No Charging
Step 3	No Charging	0.484C	0.547C	0.594C	0.688C	0.625C	0.578C	0.516C	No Charging
Step 2	No Charging	0.547C	0.609C	0.656C	0.766C	0.703C	0.641C	0.578C	No Charging
Step 1	No Charging	0.594C	0.672C	0.734C	0.844C	0.766C	0.703C	0.641C	No Charging
Step 0	No Charging	0.719C	0.797C	0.875C	1.000C	0.922C	0.844C	0.766C	No Charging

Table 2. Charging Voltage with Step Charging and JEITA

TEMPERATURE	TOO COLD	COLD2	COLD1	COOL	ROOM	WARM	HOT1	HOT2	TOO HOT
	<-10°C	-10°C – 0°C	0°C – 10°C	10°C – 20°C	20°C – 40°C	40°C – 45°C	45°C – 50°C	50°C – 55°C	>55°C
Step 4	No Charging	4.115V	4.145V	4.175V	4.200V	4.175V	4.145V	4.125V	No Charging
Step 3	No Charging	4.090V	4.120V	4.150V	4.175V	4.150V	4.120V	4.090V	No Charging
Step 2	No Charging	4.065V	4.095V	4.125V	4.150V	4.125V	4.095V	4.095V	No Charging
Step 1	No Charging	4.040V	4.070V	4.100V	4.125V	4.100V	4.070V	4.040V	No Charging
Step 0	No Charging	4.015V	4.045V	4.075V	4.100V	4.075V	4.045V	4.015V	No Charging